



A "Trip" To The ICU

Man injects magic mushroom tea into his veins, only to end up in the ICU with fungus growing inside him. <http://digit.in/feb21-15>



Lab-Cultivated Meat

Researchers develop new form of cultured artificial meat from rabbit cells, promising more natural flavour. <http://digit.in/feb21-16>

technique to search through the genes of flies with that of crustaceans, the closest relatives of insects with wings. In ground up brine shrimp and crayfish, the researchers found matching genes. The crustaceans had two genes that were known to encode wings for insects, *pda* and *apterous*. However, animals are full of anachronistic genes, genetic material that carries the necessary blueprints for building things that were no longer needed, but were once useful to a long dead ancestor. The researchers developed a method of tagging the proteins created by the *adp* and *apterous* genes with a chemical marker. When the marker was injected into shrimp larva, the area that would develop into the gills stained black.

It was only around 2010 that scientists began to appreciate the fact that the insects were more closely related to the crustaceans than the myriapod (centipedes), within arthropod groups. This was because insects were clumped with myriapods because of morphological similarities, but the scientists looking for hints on wing evolution hit a dead end.

However, when comparing the wings to crustaceans, researchers found the same genes at work in *Parhyale*, a tiny crustacean, fruitless and the *Tribolium* beetle. Using CRISPR-Cas9, researchers from Chicago systematically removed five genes responsible for the leg patterning in the insects and the crustacean. The five genes provided instructions to six leg segments farthest from the body wall. In the crustacean, they also provided instruction for a seventh leg segment, the one closest to the body wall. However, in the insects, had no such segment, so where did it go? There actually was a theory in the 1980s, according to which the insects not only absorbed the closest body segments into the body walls, but lobes on the legs turned into wings. According to the theory, when an ancestral crustacean of the insects moved to land, the leg segments closest to the body got absorbed, per-

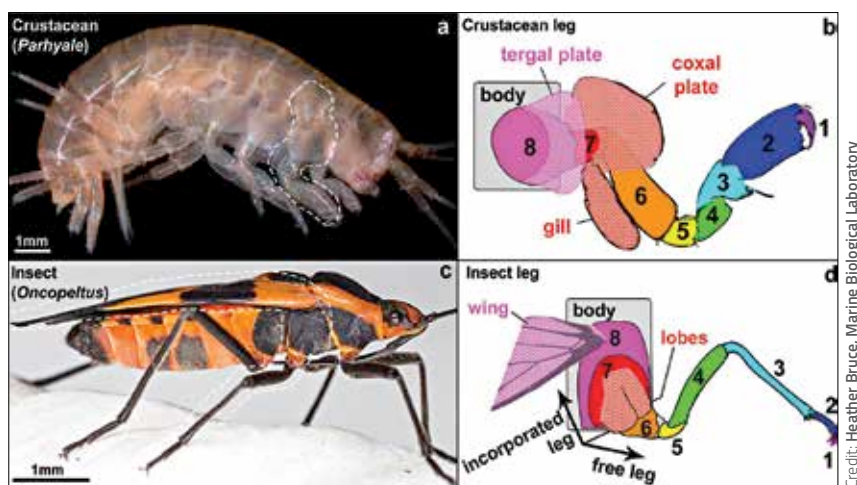
haps as a method of reducing weight in the new environment. The leg lobes then moved to the back of the insects, providing the evolutionary spring-board for flight to take off.

THE DUAL ORIGIN HYPOTHESIS

In 2010, another study conducted on gene expression in simple insects compared three genes related to wings, known as *apterous*, *wingless* and *vestigial*, on two primitive insects, mayflies and bristletails. Both these insects are considered candidates for early insect wing evolution. The researchers found that both the tergal

to evolve into novel structures is as yet unknown.

As insect bodies are segmented, and the ancestral arthropod was also a creature with simple segments and rudimentary structures such as legs. As the creatures evolved, each of the structures on the segments evolved further, such as say legs and antenna. By comparing the development and evolutionary history of the structures on one segment, with the homologous structures on the next (called homologs), scientists can potentially pinpoint the exact mechanism by which insects' wings



The segments labelled 7 and 8 were absorbed into the insect body.
The lobe on 8 later formed the wings.

margin and the pleural region shared a part with the gene network that was in action during the wing development of *Drosophila*. The finding hinted at a fusion in the schism between the theories on insect wing evolution, one where both pleural and tergal tissues are the origin of wings. This finding gave rise to a new hybrid theory, where the two otherwise unrelated tissues worked in tandem to develop the wings. The tergal tissues provided the flat blade like structure of the wings, while the pleural tissues provided the articulation mechanism, as well as the muscles. The dual origin hypothesis neatly ties up the loose ends in both theories. However, a genetic mechanism that could allow for such a merger between two tissues

evolved. Unfortunately, wings are so strange that scientists simply do not know what to compare them to. Even then a few serial homologs have been identified. Cockroaches, bristletails, treehoppers and beetles all have tergal serial homologs, while bristletails, mayflies and beetles all have pleural serial homologs. It might be possible that the wings have diverged morphologically from their serial homologs so much, that we might not even be able to recognise the structures that are the serial homologs of wings. The application of molecular biology and developmental biology together (known as the evo-devo approach) can help uncover these hidden homologs, and provide a clearer picture of insect wing evolution in the future. ■